**Table 2.1 Disaccharidases and oligosaccharidases of the intestinal brush border**

| Enzyme | Cleavage Specificity |
|---|---|
| Maltase | Maltose, maltotriose; also acts as exoglycosidase on $\alpha(1 \rightarrow 4)$ bonds at the non-reducing end of starch and starch – derived oligosaccharides |
| Lactase* | Lactose; also cellobiose# |
| Cerebrosidase* | Gluco – and galactocerebroside |
| Sucrase | Sucrose; also maltose and maltotriose |
| Isomaltase | $\alpha(1 \rightarrow 6)$ bonds in isomaltose and $\alpha$ –limit dextrins |
| Trehalase | • Trehalose |

* The lactase and cerebrosidase activities reside in two different globular domains of the same polypeptide.
# Cellobiose is a disaccharide of two glucose residues in $\beta(1 \rightarrow 4)$ glycosidic linkage.
• Trehalose is a disaccharide with a structure of ∝-D-glucopyranosy l -∝ -D-glucopyranoside (in 1,1 glycosidic linkage); common only in mushrooms.

## Absorption of carbohydrates

- Only monosaccharides can be absorbed by the villi. However, few disaccharides are taken up by pinocytosis and later hydrolysed to monosaccharides.
- The principal monosaccharides that are present in the diet or formed as end products of digestion are glucose (80%), galactose, fructose and pentoses like ribose, xylulose and arabinose.
- The absorption of monosaccharides mainly takes place in the duodenum and the upper part of the small intestine.
- The order of absorption of important monosaccharides are :
  Galactose > Glucose > Fructose > Mannose > Pentoses
- Monosaccharides are absorbed by both passive diffusion as well as active transport. Different monosaccharides are absorbed in different ways. Glucose and Galactose are the prime monosaccharides that are absorbed actively, while fructose and pentoses are absorbed passively by facilitated diffusion.

## Facilitated diffusion

Initially, when the glucose concentration in the digested food is very high, facilitated diffusion plays a major role in absorption. Since glucose is a highly polar molecule, glucose is absorbed passively by a transporter, GLUT2 (Glucose Transporter 2). Fructose is absorbed by GLUT5 transporter in a passive manner.



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